

N-Tidal Study – Vyom Khanna

Total tests	8,678
Positive diagnosis rate	21.2% (1,839 tests)
Overall error rate	17.9% (1,552 tests affected)
Practices with issues	2 (PRC-007 and PRC-009)
Handsets with issues	3 (handset-00005, 00013, 00014)
Faults identified	6 (2 critical, 2 high, 1 medium, 1 low)

Overall

The study is running well across 8 of 10 practices. Test volumes are stable, error rates are normal, and diagnostic results are consistent. Two practices need attention before the study concludes.

Summary

Issue	Practice or Device	Status	Action needed
Device faults causing high errors	PRC-007	Urgent	Retrieve handsets, review all results
Handsets not updated to latest software	Handsets 13 and 14	Urgent	Update once connectivity is restored
Weekend testing outside study plan	PRC-009	Review	Contact practice, clarify protocol
Health data missing for one handset	Handset-00005 at PRC-003	Review	Raise support ticket
Intermittent signal alerts	Handsets 5 and 6	Monitor	No action needed yet
3 incomplete device log entries	Handsets 6, 15, 18	Low	Flag for engineering team

Note on Data Quality

Before analysis the hardware telemetry data was checked and cleaned. 33 duplicate entries were removed, 8 records with empty data were dropped, and 102 records with missing device labels were recovered using their message content. These issues were present in the raw data feed and did not change the study findings. They have been flagged to the engineering team for review.

The test records table was fully complete with no missing values or duplicates.

Study Progress

Test volumes

8,678 tests were completed across 10 practices over 30 days. On weekdays all practices ran a similar number of tests each day, averaging around 377 tests across the group. There were no gaps or drops in activity that would suggest practices stopped participating.

Diagnostic outcomes

21.2% of tests returned a positive result. This rate is consistent across all practices, ranging from 19% to 23%, and is in line with what would be expected for a GP patient population being tested for a respiratory condition.

Error rates

The overall error rate across all tests is 17.9%. Most of this comes from user errors such as timeouts (5%) and technique errors (3%), and from software checks (6%). These rates are within acceptable bounds for a field study of this type. One practice, PRC-007, is a significant exception and is covered in the next section.

Issues Requiring Action

Urgent: PRC-007 device problems

PRC-007 has an error rate of nearly 50%, compared to around 13% for all other practices. The two handsets at this practice have hardware faults and did not receive the latest software update. All results from this practice should be reviewed before being used clinically.

What is happening at PRC-007

The two handsets at PRC-007 (handset-00013 and handset-00014) have been logging persistent device health alerts throughout the study. Handset-00013 in particular has a battery voltage that is consistently lower than normal, around 3.38 volts compared to a healthy level of 3.81 volts. Low battery voltage in diagnostic devices causes the sensors to drift, which produces incorrect readings and triggers calibration errors.

Both handsets also have very poor wireless signal, around 6 out of a normal 22. This poor connection likely prevented them from receiving the firmware update released on 28 April. All other 18 handsets received this update. After that date the error rate at PRC-007 continued to climb while all other practices improved.

What needs to happen

- Retrieve handsets 00013 and 00014 from PRC-007 for physical inspection and battery testing.
- Consider issuing replacement handsets to PRC-007 so testing can continue.
- Mark all diagnostic results from PRC-007 for clinical review before they are used in any analysis or patient decisions.

For review: PRC-009 weekend testing

PRC-009 has been running patient tests on weekends. This was not part of the agreed study plan. The 280 weekend tests look real but they fall outside the period when TidalSense technical support is available, which means any device fault over a weekend would have gone unaddressed.

What needs to happen

- Contact PRC-009 to understand whether weekend clinics are intentional.
- If weekend testing will continue, update the study protocol and arrange weekend support cover.
- If weekend testing was not approved, ask the practice to stop.

Other Findings

Handset-00005 stopped sending health data

One handset at PRC-003 (handset-00005) stopped transmitting device health logs after 15 April but continued to run tests for the rest of the study. This means there is no device health data for this handset for the second half of the study, and we cannot confirm which software version it was running. A support ticket should be raised. The tests it completed are not necessarily invalid, but they cannot be verified against device health data.

Signal and battery alerts on other handsets

Two further handsets (handset-00005 and handset-00006, at different practices) have had intermittent low signal alerts during the study. These appear to be brief connectivity drops rather than a persistent fault. Both handsets are otherwise performing normally. They should be monitored for the remainder of the study but no immediate action is needed.